

Jatin Chowdhury

Audio Signal Processing Engineer

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Education

Massachusetts Institute of Technology

Cambridge, MA

PH.D IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Sept. 2025 - PRESENT

- Relevant research: real-time circuit simulation, audio software development, signal processing.
- Relevant coursework: numerical algorithms.

Stanford University, Center for Computer Research in Music and Acoustics

Palo Alto, CA

M.A. IN MUSIC, SCIENCE, AND TECHNOLOGY

Sept. 2018 - June 2020

- Relevant coursework: advanced digital signal processing, machine learning, spatial audio, music recording.
- Research: nonlinear signal processing, virtual analog modelling, modal signal processing.

University of Southern California

Los Angeles, CA

B.S. IN ELECTRICAL ENGINEERING, PHYSICS (MINOR), MUSIC RECORDING (MINOR)

Aug. 2014 - May 2018

- Relevant coursework: signal processing, circuit design, digital logic, electromagnetics, software design.

Experience

Chowdhury DSP

Renton, WA

AUDIO SIGNAL PROCESSING ENGINEER

Oct. 2021 - PRESENT

- Developed signal processing algorithms for real-time audio processing and synthesis.
- Contributed to open-source audio plugins and libraries.

Tesla Motors

Palo Alto, CA

AUDIO TEST ENGINEER

Aug. 2020 - Oct. 2021

- Developed audio system signal flow layouts for vehicles using AudioWeaver.
- Developed end-of-line tests to ensure vehicle audio system quality using Python and C++.
- Contributed to the signal processing and tuning for the vehicle pedestrian warning speaker.

Publications & Presentations

Performance Engineering for Modern DSP

September 2025

SIGNAL PROCESSING SUMMIT

Milpitas, CA

- Discussed performance engineering based on modern advancements in hardware technology.
- Discussed ideas for teaching novice engineers about performance.

Inference-Time Structured Pruning for Real-Time Neural Network Audio Effects

Sept. 2025

PROC. OF THE 28TH INTERNATIONAL CONFERENCE ON DIGITAL AUDIO EFFECTS

Ancona, IT

- Co-authored with Christopher Johann Clarke.
- Available on the DAFx Archives.

Emulating Diode Circuits with Differentiable Wave Digital Filters

June 2022

19TH SOUND AND MUSIC COMPUTING CONFERENCE

Saint Etienne, France

- Co-authored with Christopher Johann Clarke.
- Available on Zenodo.

RTNeural: Fast Neural Inferencing for Real-Time Systems

June 2021

ARXIV E-PRINTS: AUDIO AND SPEECH PROCESSING

Online

- Presents a neural network inferencing library for real-time systems.
- Available on the ArXiv.

Real-Time Physical Modelling for Analog Tape Machines

Sept. 2019

PROC. OF THE 22ND INTERNATIONAL CONFERENCE ON DIGITAL AUDIO EFFECTS

Birmingham, UK

- Oral presentation at the DAFx-2019 conference.
- Available on the DAFx Archives.

Skills

Programming Languages

C/C++, Python, Bash, Jai, MATLAB, Faust, LaTeX

Tools/Frameworks

CMake, Git, Linux CLI, Visual Studio, Xcode, GitHub Actions, JUCE API, CLAP